

# **Temitope Ologunbaba O.**

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## **EDUCATION**

**Federal University of Technology, Akure, Nigeria.**

**2015 – 2021**

Bachelor of Engineering; Electrical/Electronics Engineering

**Second Class Upper (3.49/4.00)**

## **SKILLS AND CERTIFICATIONS**

**Certifications:** Intermediate Machine Learning, IBM - Introduction to Artificial Intelligence, DataCamp - Intermediate Python, Coursera - Deep Learning, Lagos Africa AI Bootcamp, Fundamentals of Grant Writing.

**Programming Languages:** Python (PyTorch, Numpy, Flask, Keras), Dart (Flutter), OpenDSS, PLC Programming, Java (Spring, Mobile), JavaScript (NodeJS), Firebase and Git, Docker, MongoDB, MySQL.

**Softwares:** MATLAB, PSpice, AutoCAD Pro, TensorFlow, and OpenCV, Microsoft Office.

**Relevant Courses/Skills:** Electrical Panel Design, Deductive Mathematics and Algebra, Inverter System Design & Integration, Technical Proficiency in Inverter and Battery Systems, Software Development, Engineering Analysis and Modeling using MATLAB, Machine Learning using Scikit-Learn, Power Analysis.

## **RESEARCH INTERESTS**

Power Systems, Machine Learning and Artificial Intelligence, Power Electronics, Communication Systems, Electronics, Robotics, Automation, Optimization, Software and Computer Engineering.

## **RESEARCH EXPERIENCE**

**Thesis: Effect of Weather Conditions on Free Space Optics, Review Paper**

**2020 - 2021**

- Built a sound transmitter and receiver using microcontroller and electronics components that transmit sound signals with only 5% attenuation within a free space.
- Simulated rain conditions to analyze the impact on Free Space Optics (FSO) transmission, measuring signal degradation using an oscilloscope.
- Reviewed and documented weather effects on FSO performance, providing insights into atmospheric interference in the City.

**Communications Laboratory - Student Researcher, FUTA**

**2018 - 2021**

- Conducted experiments and simulations on Optical systems, improving transmission efficiency.
- Analyzed network protocols and signal processing techniques, leading to a deeper understanding of communication theory.
- Collaborated with research teams, contributing to the development of new models and technologies.

## **PROFESSIONAL EXPERIENCE**

**Greenpeg Engineering - Power System Engineer**

**November 2022 – Present**

- Led the development of power distribution solutions, improving energy conversion efficiency by 25% across residential, commercial, and industrial sectors.
- Designed inverter systems through load analysis, achieving a 20% increase in energy balance between battery and grid systems, optimizing energy supply.
- Managed installation and commissioning of systems, ensuring 100% operational success from concept to implementation, reducing downtime and improving performance.
- Implemented maintenance protocols, extending product life by 15% and reducing system downtime by 20%, ensuring consistent peak performance.

**Research/Development Engineer (Remote), Blumefy**

**September 2021 – October 2022**

- Conducted app research to gather insights into user behaviors, needs, and pain points, then analyze to work with the results.

- Designed, prototyped, developed and tested new features, increasing feature engagement by 15% and driving innovation for future app releases.
- Optimized app performance and resolved technical issues from customer support, while reducing app crashes by 30% and significantly improving app stability and user satisfaction.

#### **Research/Development Engineer, Komodo (Part-time).**

**September 2019 – Present**

- Specialized in designing, developing desktop and mobile applications using Java, Flutter and Dart, and developed over 30 applications both on play store and app store for the clients.
- Deployed the backend application to GCP server. I also managed the server, fixed server bugs, automated our processes and increased stability to up to 99% uptime.

#### **Class Teacher, Mariam Joseph Anglican Grammar School (Service Corp)**

**March 2022 – Feb. 2023**

- Designed engaging science lessons in advanced mathematics and physics, increasing student comprehension by 30% through hands-on experiments and interactive learning activities.
- Led classroom discussions and projects, fostering critical thinking and problem-solving skills, and boosting student participation by 40%.
- Conducted physics lab sessions and hands-on activities, enhancing student understanding through practical experiments, and increasing performance through tailored assessments and guided support.

#### **System Engineer Intern, National Meteorological Agency**

**June 2019 – December 2019**

- Maintained and Developed Desktop and System Solutions for the agency as a junior Intern.
- Weekly technical maintenance of airport runway electrical instruments on the field.
- Collection of weather data and working with meteorologists in analyzing data, and interpreting them.

### **ACADEMIC AND PROFESSIONAL ACHIEVEMENTS**

- Speaker, **Lagos Machine Learning Bootcamp.** **2024**
- Recipient, **Annual Agbami Scholarship Award.** **2017 - 2021**
- 3<sup>rd</sup> Best Graduating student in the Department of EEE, FUTA. **2021**
- Recipient, **Engineers-In-Society Symposium Certificate.** **2017**
- 4x Dean's/Scholars' List, FUTA. **2016 - 2018, 2021**
- Top 3 Best graduating students in the Department of Electrical Engineering, FUTA. **2022**
- Top 5, **Read to Lead Africa Scholarship Initiative.** **2018**
- Recipient, **National Merit Scholarship** – Awarded for exceptional academic performance in undergraduate studies. **2018**
- Winner, **S.U. Writing, FUTA** – Recognized for outstanding writing on social issues. **2015**
- Recipient, **Impact Leaders Club Mentor Award** – Recognized for exceptional mentorship and leadership in guiding students. **2019**
- Honored Member, **Future Engineers of Africa.** **2023**

### **CONFERENCES ATTENDED**

- **Africa Renewable Energy Forum (AREF)** **2023**
- Need-driven engineering research for entrepreneurial development in developing countries. **2019, 2021**
- **Society of Engineers**, Virtual International conference. **2024**

### **PROFESSIONAL AFFILIATIONS**

- Member, **IEEE(Institute of Electrical and Electronics Engineers, Nigeria Chapter)** **2019 - Present**
- Member, **Future Engineers of Africa.** **2023**
- Member, **Nigerian Universities Engineering Students Association, FUTA.** **2015 – 2021**

- Graduate Member, **Nigerian Society of Engineers.** 2021 – Present
- Member, **Impact Your World Leadership Initiative.** 2018 – Present
- Member, **Impact Leaders Club, FUTA.** 2015 – 2021

## ARTICLES AND PUBLICATIONS

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- **Effect of Weather Conditions on Free Space Optics**, Review Paper (Ongoing)
- **Building a Movie Recommendation System Using Content-Based and Collaborative Filtering.**
- **Harnessing AI in Mobile Apps: Implementation with Flutter.**
- **The Mathematics of Neural Networks.**
- **Building an Image Recognition System with Python, TensorFlow, and Keras.**
- **Mastering Conversational AI: A Deep Dive into Natural Language Understanding for Chatbots.**
- **Comprehensive Guide to Using Docker with Python: Focus on Machine Learning Applications.**
- **Unleashing the Power of Neural Networks in Artificial Intelligence.**

## PROJECTS

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- **Transmitter/Receiver System**  
This project explores the use of electrical components to build a transmitter which sends sound signals and a receiver that can be tuned to different frequencies to capture the signal, it was also used to measure the attenuation of the signal in face of different weather conditions that might affect it.  
**Technologies: Free Space Optics.**
- **Heart Disease Predictor** [REPO LINK](#)  
This project explores how to build a model capable of predicting (up to 95%) heart disease whether or not an individual has heart disease based on their medical attributes. **Technologies: Python, Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn.**
- **CNN implementation for MNIST Digit Classification** [REPO LINK](#)  
This project implements a Convolutional Neural Network (CNN) to classify handwritten digits from the MNIST dataset. The model is designed to achieve an accuracy of 99.5% or more by adding a single convolutional layer and a single MaxPooling 2D layer to the architecture. The goal is to demonstrate the effectiveness of CNN image classification tasks. **Technologies: Python, Tensorflow.**
- **3rd Party Privacy Detector** [REPO LINK](#)  
This is a real time camera detection system I built to detect if a device is about to be snapped from an external camera and was built with tensorflow. I built the model with Tensorflow and it was deployed on a live mobile app. **Technologies: Java, Flutter, Dart, Python, Tensorflow, Keras.**
- **Movie Recommendation System** [REPO LINK](#)  
This project demonstrates how to build a content-based movie recommender system, allowing you to manage dependencies and deploy it in various environments. The recommender system is built using Python and utilizes TF-IDF and cosine similarity to recommend movies based on genre similarities. **Technologies: Java, Flutter, Dart, Python, Docker, Scikit-learn, Tensorflow, Cosine Algo.**

## LEADERSHIP AND VOLUNTEERING EXPERIENCE

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- **General Secretary**, NIEEES, FUTA. 2019 – 2021
- **Speaker**, Lagos Machine Learning Bootcamp. 2024
- **Class Teacher**, Mariam Joseph Grammar School, National Youth Service Corp (NYSC). 2022
- **Student Mentor**, Impact Leaders Club. 2022
- **Undergraduate Tutor/Sound, Light and Power Technician**, CACCF, FUTA. 2015 - 2019
- **Course Representative**, Department of Electrical/Electronics Engineering. 2016 - 2018
- **Community Outreach**, Reading Awareness Society for Development in Africa(RASDA) 2014
- **Participant**, Reading Awareness Society for Development in Africa(RASDA) Volunteering 2014